

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Mr. Rawson • GCSE Computer Science



Year 10 2026

# Binary Addition

Work in pencil. One column at a time, right to left — the 1s column first. A carry is written in the **next column to the left**, because that is the column it gets added into. In the four rules below, a **ringed** digit is a carry.

## The four rules

CARRY

|  |          |
|--|----------|
|  | 0        |
|  | 0        |
|  |          |
|  | <b>0</b> |

$$0 + 0 = 0$$

CARRY

|  |          |
|--|----------|
|  | 0        |
|  | 1        |
|  |          |
|  | <b>1</b> |

$$0 + 1 = 1$$

CARRY

|          |          |
|----------|----------|
|          | 1        |
|          | 1        |
|          |          |
| <b>①</b> | <b>0</b> |

$$1 + 1 = 0 \text{ carry } 1$$

CARRY

|          |          |
|----------|----------|
|          | 1        |
|          | 1        |
|          | <b>①</b> |
| <b>①</b> | <b>1</b> |

$$1 + 1 + 1 = 1 \text{ carry } 1$$

## Worked example 43 + 74

| Place value | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | Denary |
|-------------|-----|----|----|----|---|---|---|---|--------|
| Number      | 0   | 0  | 1  | 0  | 1 | 0 | 1 | 1 | 43     |
| Number      | 0   | 1  | 0  | 0  | 1 | 0 | 1 | 0 | 74     |
| Answer      | 0   | 1  | 1  | 1  | 0 | 1 | 0 | 1 | 117    |
| Carries     |     |    |    | 1  |   | 1 |   |   |        |

Read it right to left. In the 2s column  $1 + 1 = 0$  carry 1, so a **1** goes in the Carries row under the 4s column and is added in there. The denary column is the check:  $43 + 74 = 117$ , and the answer row reads 117.

Name: \_\_\_\_\_

Date: \_\_\_\_\_

**Your turn**1. Write **12** and **7** into the grid in binary, then add them.

| Place value | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | Denary |
|-------------|-----|----|----|----|---|---|---|---|--------|
| Number      | 0   | 0  | 0  | 0  | 1 | 1 | 0 | 0 | 12     |
| Number      | 0   | 0  | 0  | 0  | 0 | 1 | 1 | 1 | 7      |
| Answer      | 0   | 0  | 0  | 1  | 0 | 0 | 1 | 1 | 19     |
| Carries     |     |    |    | 1  | 1 |   |   |   |        |

2. Write **46** and **67** into the grid in binary, then add them.

| Place value | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | Denary |
|-------------|-----|----|----|----|---|---|---|---|--------|
| Number      | 0   | 0  | 1  | 0  | 1 | 1 | 1 | 0 | 46     |
| Number      | 0   | 1  | 0  | 0  | 0 | 0 | 1 | 1 | 67     |
| Answer      | 0   | 1  | 1  | 1  | 0 | 0 | 0 | 1 | 113    |
| Carries     |     |    |    | 1  | 1 | 1 |   |   |        |

Name: \_\_\_\_\_

Date: \_\_\_\_\_

3. Write **89** and **52** into the grid in binary, then add them.

| Place value | 128      | 64       | 32       | 16       | 8        | 4        | 2        | 1        | Denary     |
|-------------|----------|----------|----------|----------|----------|----------|----------|----------|------------|
| Number      | <b>0</b> | <b>1</b> | <b>0</b> | <b>1</b> | <b>1</b> | <b>0</b> | <b>0</b> | <b>1</b> | 89         |
| Number      | <b>0</b> | <b>0</b> | <b>1</b> | <b>1</b> | <b>0</b> | <b>1</b> | <b>0</b> | <b>0</b> | 52         |
| Answer      | <b>1</b> | <b>0</b> | <b>0</b> | <b>0</b> | <b>1</b> | <b>1</b> | <b>0</b> | <b>1</b> | <b>141</b> |
| Carries     | <b>1</b> | <b>1</b> | <b>1</b> |          |          |          |          |          |            |

4. The two binary numbers are already in the grid. Add them, then fill in the denary column as a check.

| Place value | 128      | 64       | 32       | 16       | 8        | 4        | 2        | 1        | Denary     |
|-------------|----------|----------|----------|----------|----------|----------|----------|----------|------------|
| Number      | 0        | 1        | 0        | 1        | 1        | 0        | 1        | 0        | <b>90</b>  |
| Number      | 0        | 0        | 1        | 0        | 1        | 1        | 0        | 1        | <b>45</b>  |
| Answer      | <b>1</b> | <b>0</b> | <b>0</b> | <b>0</b> | <b>0</b> | <b>1</b> | <b>1</b> | <b>1</b> | <b>135</b> |
| Carries     | <b>1</b> | <b>1</b> | <b>1</b> | <b>1</b> |          |          |          |          |            |

Name: \_\_\_\_\_

Date: \_\_\_\_\_

5. Write **107** and **93** into the grid in binary, then add them.

| Place value | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | Denary |
|-------------|-----|----|----|----|---|---|---|---|--------|
| Number      | 0   | 1  | 1  | 0  | 1 | 0 | 1 | 1 | 107    |
| Number      | 0   | 1  | 0  | 1  | 1 | 1 | 0 | 1 | 93     |
| Answer      | 1   | 1  | 0  | 0  | 1 | 0 | 0 | 0 | 200    |
| Carries     | 1   | 1  | 1  | 1  | 1 | 1 | 1 |   |        |

6. Write **150** and **130** into the grid in binary, then add them.

| Place value | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | Denary |
|-------------|-----|----|----|----|---|---|---|---|--------|
| Number      | 1   | 0  | 0  | 1  | 0 | 1 | 1 | 0 | 150    |
| Number      | 1   | 0  | 0  | 0  | 0 | 0 | 1 | 0 | 130    |
| Answer      | 0   | 0  | 0  | 1  | 1 | 0 | 0 | 0 | 24     |
| Carries     |     |    |    |    | 1 | 1 |   |   |        |

**6(b)** Check your answer against the denary column. Something has gone wrong. What is this called, and why has it happened? **Overflow.**  $150 + 130 = 280$ , and **280 needs nine bits.** Only eight columns exist, so the 1 that should sit in a 256s column has nowhere to go and is lost — the grid reads 24, which is  $280 - 256$ . The processor would set an overflow flag.